A cell-type-weighted regulatory prior resolves the transcription factors driving bifurcations in a cross-study *Arabidopsis* irradiation pseudo-time-series

Richard J. Barker*
TODO: co-author list

August 29, 2026

Abstract

The Dynamic Regulatory Events Miner (DREM) reconstructs when transcription factors (TFs) act by finding bifurcations in time-series expression and attributing each to a regulator, using a static TF–gene interaction prior [1, 2]. That prior is normally binary and tissue-blind: an edge either exists or it does not, regardless of whether the TF is expressed in the cells producing the signal. We show that a single-cell atlas can be made to act on the model itself rather than on its annotation, by writing cell-type context into the score column DREM already reads. Screening all 62 *Arabidopsis thaliana* studies in NASA's Open Science Data Repository (OSDR), we assemble a 10-point γ-irradiation pseudo-time-series spanning 10 min to 72 h from 108 samples, and a second 3-point spaceflight series from 115 samples. We deconvolve each timepoint against a 183-cluster single-nucleus atlas, weight 976,588 DAP-seq and ChIP-seq TF–gene edges by the cell types actually present, and compare the resulting models against flat and binding-only priors under an otherwise identical search. The wild-type/*sog1-1* design provides an internal positive control: the canonical DNA-damage response is induced across all 6 sentinel genes in wild type and collapses in the mutant. Turning the two-genotype contrast into a decoder and scoring it across every *Arabidopsis* study in the repository recovers 11 of 12 labelled irradiation contrasts with 1 false positives, ordered by dose, and finds no other OSDR plant study carrying the signature. No DREM transcription factor is altered in spaceflight and no flight contrast matches any phase of the radiation trajectory — but the assay's diagnostic arm needs 40 cGy while ISS missions deliver 0.39–2.485 cGy, so that null bounds the effect rather than excluding damage. Of 6 bifurcations in the wild-type model, 3 change their top-ranked regulator with the prior — but not the canonical ones: SOG1 ranks first of 478 TFs in wild type under every prior and falls to rank 574 in *sog1-1*. Attribution moves where the evidence was marginal and holds where it was strong.

1 Introduction

Time-series transcriptomics answers *when* a response happens, but not *who* drives it. The Dynamic Regulatory Events Miner (DREM) closes that gap by fitting an input–output hidden Markov model in which co-expressed genes travel together until a *bifurcation*, and by attributing each bifurcation to the transcription factors (TFs) whose targets are enriched along one branch [1]. DREM 2.0 added model-selection and scoring improvements [2], and iDREM extended the framework to multi-omic inputs and interactive exploration [3].

The attribution step depends entirely on a second input: a static TF–gene interaction table. In practice that table is binary and tissue-blind. An edge asserts that a TF *can* bind a promoter, not that the TF is present in the cells generating the measured signal. For a bulk time-series

*Center of Space Exploration (CoSE). Correspondence: dr.richard.barker@gmail.com

drawn from a whole organism this is a real limitation, because the transcriptome is a mixture and the mixture changes.

Single-cell data is the obvious remedy, and iDREM already accepts it — but only after the fact. Its documentation is explicit that the cell-type data "is not used when predicting the iDREM model"; it annotates finished nodes rather than informing them. The contribution of this work is to move that information upstream, into the model itself, without modifying the engine. DREM reads the third column of its TF–gene file as a score in $[0, 1]$ rather than as a binary flag, and that column is a sufficient channel: an edge whose TF is absent from the responding cell types can be down-weighted before the search begins.

We test this on *Arabidopsis thaliana* data from NASA's Open Science Data Repository (OSDR). Screening all 62 OSDR *Arabidopsis* studies for compatible genotype and treatment metadata yields two cross-study pseudo-time-series. The first is a γ-irradiation series of 10 timepoints spanning 10 min to 72 h. Its core — OSD-498, OSD-508 and OSD-510 — is a re-assembly of one experiment deposited under three accessions, that of Bourbousse et al. [4], who built their own DREM model over these data and reported 11 co-expressed gene groups. That gives this work an unusual advantage: a published DREM analysis of the same material to calibrate against, and a genetic positive control, since the series contains both wild type and *sog1-1*, a knockout of the NAC master regulator of the plant DNA-damage response [5]. The second series follows spaceflight seedling development across 3 timepoints and tests whether the method generalises to a different stress.

2 Methods

2.1 Study selection

All *Arabidopsis thaliana* studies in OSDR were retrieved through the `biodata` v2 metadata API, returning sample-level factor values for 62 studies. Studies were admitted to a cohort only where a numeric time axis could be parsed from a factor-value field and the assay was Illumina RNA-seq. Cohort A (γ-irradiation) comprises 5 studies and cohort B (spaceflight development) 4. 7 studies were excluded with a recorded reason (`scripts/cohorts.py`).

Two exclusions matter. OSD-219 re-deposits all 32 of OSD-218's BioSample identifiers verbatim; including both would double-count every sample. This is re-detected each run from sample-name overlap rather than hard-coded, so the exclusion cannot outlive the condition that justifies it. The microarray irradiation studies (OSD-46, OSD-320, OSD-329) were held out of the primary analysis because pooling them with RNA-seq would confound platform with timepoint.

2.2 Pseudo-time-series construction

Raw RSEM counts and GeneLab runsheets were retrieved from the OSDR file API. Counts were converted to CPM and $\log_2$-transformed *within* each study, then expressed as a $\log_2$ ratio against that study's own untreated controls. This within-study referencing removes most between-study offset by construction, because a per-study additive term cancels between numerator and denominator. A residual per-study offset was then fitted as the median gene-wise difference from the cross-study consensus, computed only at the 7 timepoints more than one study measured; a study sharing no timepoint with any other is left unadjusted rather than shifted against a reference it never overlaps.

Because every value is already a ratio against the untreated state, an explicit $t = 0$ column of zeros is prepended and DREM is run with `No normalization/add 0`. Omitting either would make DREM re-anchor on the 10-min timepoint, which is already strongly induced, and the entire early response would vanish from the model.

OSD-782 is held out rather than pooled. It is an independent laboratory *and* a different dose regime (10–100 cGy against the core's high-dose γ), so fitting it a shared offset would remove real biology.

Batch correction was verified against six canonical DNA-damage-response genes (BRCA1, RAD51, PARP2, SMR7, GMI1, XRI1). The pipeline fails if fewer than three remain 2-fold responsive in the wild-type arm — the check exists to catch a correction that has flattened the signal it was meant to preserve.

2.3 TF–gene prior

A binding prior was built from the *Arabidopsis* DAP-seq cistrome [6], retrieved as GEO GSE60143. Of 814 peak files, 194 name the assayed TF by AGI locus and 557 by gene symbol; symbols were resolved through a map built from the Ensembl Plants release-54 annotation, TAIR gene aliases [7] and ConnecTF [8]. 63 files remain unresolved (ChIP-seq controls, non-*Arabidopsis* orthologue assays, and post-2014 symbols) and are listed in the run report. Resolving symbols is not cosmetic: the unresolved set is dominated by the NAC, MYB, WRKY and bZIP families, which is where the DNA-damage regulators are.

Peaks were assigned to genes within 1 kb of a TSS using the same Ensembl release-54 TAIR10 annotation GeneLab used to produce the count matrices, so binding and expression share one coordinate system and one locus vocabulary. Edge scores were rank-normalised within each TF, so a score encodes binding confidence relative to that TF's other sites rather than its sequencing depth. The prior contains 976,588 edges over 478 TFs, with a median of 2,467 targets per TF.

SOG1. DAP-seq does not assay SOG1, so the DAP-seq prior cannot attribute any bifurcation to the very regulator this cohort is about. SOG1 edges were therefore called from the ChIP-seq of the same experiment (OSD-496 / GEO GSE112529), in which SOG1-3xFLAG was immunoprecipitated 20 min and 1 h after irradiation. OSDR holds only raw reads for OSD-496, so coverage was taken from GEO. A 200 bp bin was retained where IP exceeded *both* its matched input and the no-FLAG wild-type IP two-fold, which removes chromatin-accessibility and antibody-background artefacts that either control alone would miss. This yielded 3,310 and 3,101 peaks at the two timepoints, and 1,720 target genes after the same TSS-proximity assignment used for DAP-seq.

2.4 The auto-decoder lever

Cell-type context comes from an auto-decoder trained on the GSE226097 single-nucleus *Arabidopsis* atlas [9], which provides 4,000 signature genes across 183 clusters and a 32-dimensional latent code per cluster.

Each timepoint's mean linear CPM profile was deconvolved against the signature matrix by non-negative least squares, with both sides scaled to unit norm and the atlas signatures returned to linear space beforehand. Deconvolution uses absolute expression, never the log-ratio series: a fraction is a statement about composition, and a fold-change has already divided composition out. 3,834 signature genes matched, with a median relative residual of 0.3388.

Writing $\bar{f}_c$ for the timepoint-averaged fraction of cluster c and $\hat{e}(g, c)$ for gene g's cluster profile scaled to its own maximum, the context of a gene is $\mathrm{ctx}(g) = \sum_c \bar{f}_c \, \hat{e}(g, c)$, and each edge is re-weighted as

$$w(\mathrm{TF}, g) \;=\; \mathrm{binding}(\mathrm{TF}, g) \;\cdot\; \mathrm{ctx}(\mathrm{TF})^{\alpha} \;\cdot\; \mathrm{ctx}(g)^{\beta},$$

with $\alpha = 1$ and $\beta = 0.5$: the TF side dominates because DREM's attribution asks which regulator explains a split, not which target. Weights are renormalised to $[0, 1]$ within each TF and floored at 0.05 where binding evidence exists, so context can down-weight an edge but never silently delete it.

Genes outside the atlas signature set are, by construction, genes whose expression varies little across the 183 clusters; their context is therefore close to uniform and their edges are passed through unchanged. 232,931 edges (24%) are atlas-informed and 16% move by more than 0.01.

2.5 Radiation signature and decoder

Gene sets were derived from the two-genotype arms. A gene is *SOG1-dependent* when it reaches $|\log_2 \mathrm{FC}| \geq 1$ in wild type, is a SOG1 ChIP target, and its *sog1-1* amplitude is below 40% of its wild-type amplitude — a genetic criterion, not a correlation with exposure. The contrast set (*SOG1-independent*) holds induced SOG1 targets that respond in both genotypes.

Each OSDR study was scored on every contrast its metadata supports: for each factor, any level matching a control vocabulary (ground control, non-irradiated, mock, untreated, zero dose) defines the reference and every other level of that factor is scored against it. Factors describing what a sample *is* rather than what was done to it — organ, tissue, age, ecotype — are excluded. Genes are ranked by $\log_2$ fold change and a set's mean normalised rank is compared with 2,000 size-matched random sets from the same ranking, giving a z-score and an empirical p. Ranks, rather than fold-change thresholds, keep the score comparable across platforms and sequencing depths.

The decision index is the conjunction $\min(z_{\text{SOG1-dep}}, -z_{\text{MYB3R-rep}})$ at a fixed threshold of 1.96, so both arms must be individually significant. The threshold is statistical rather than fitted: a threshold placed between the labelled classes is hostage to OSDR's positives including a 10 cGy exposure that produces no response.

2.6 Tissue attribution

Tissue association was tested against the atlas expression matrix directly, not through the deconvolution. Each gene's cluster profile is scaled to its own maximum, a set's mean scaled expression per cluster is compared with 1,000 size-matched random sets, and the result is a per-cluster z and empirical p. Deconvolution stability was measured as the median relative change in fraction between adjacent timepoints, restricted to compartments carrying at least 2% mass, and the time-averaged composition was bootstrapped over timepoints.

2.7 Spaceflight TF activity, statistics and machine learning

TF activity is the rank statistic above applied to each transcription factor's target set (edges scoring ≥ 0.5, at least 20 targets), giving 474 features per contrast. The null is analytic rather than permuted: for a size-n subset of N ranks the subset mean has variance $\frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}$, the finite-population correction, which is exact. Agreement with permutation was verified across set sizes from 20 to 3,000 (`-verify-null`); differences were within Monte-Carlo error.

Per-TF tests average contrasts within a mission before testing, so the unit of analysis is the mission. Flight effects are one-sample t tests over mission means; flight-versus- radiation uses Welch's t; both are Benjamini–Hochberg corrected across TFs.

Classification is L2-regularised logistic regression ($C = 0.1$, balanced classes) and a random forest, under leave-one-mission-out cross-validation, with radiation studies each forming their own group. Significance comes from permuting the group-to-label assignment, not the contrast labels. A platform classifier trained on identical features under identical cross-validation serves as the confound control.

Trajectory projection correlates each contrast's TF-activity vector (Spearman) against the same vector computed at each DREM timepoint. Because an argmax over eight profiles always exists, the reported statistics are the shape of the mean correlation profile and the fraction of per-contrast argmaxes falling at or beside its peak.

2.8 Model fitting and comparison

Models were fitted with DREM 2.0.7 [10] in batch mode (`java -jar drem.jar -b`), with a pinned random seed, node penalty 40, at most three paths out of a split, and a $|\log_2| \geq 1$ expression filter. Replicates were supplied as separate full matrices rather than pre-medianed.

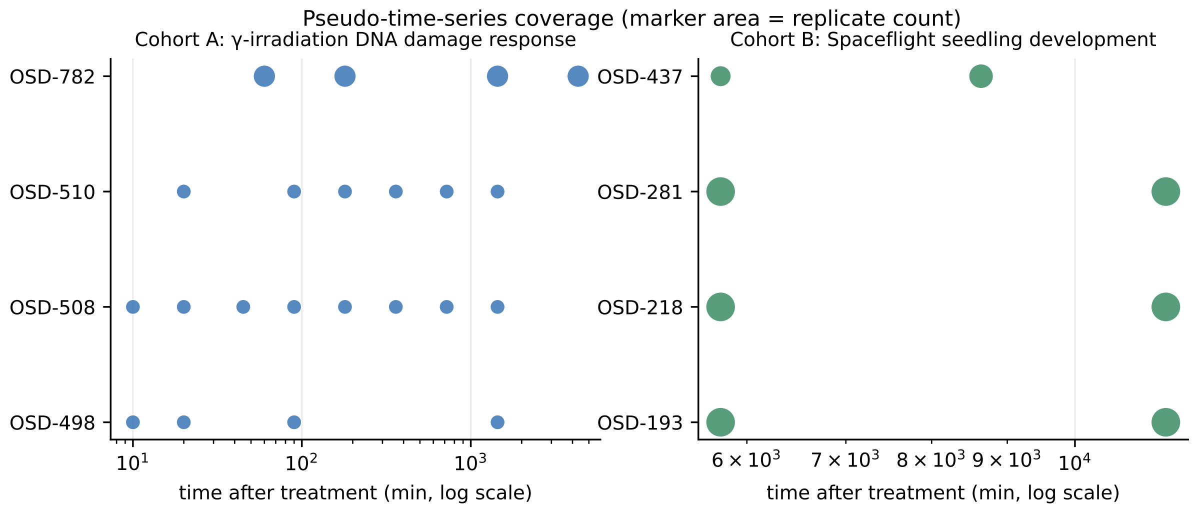

Figure 1: Pseudo-time-series coverage. Each marker is a timepoint measured by one accession; marker area is the replicate count. Cohort A's accessions overlap at 7 timepoints, which is what allows a residual between-study offset to be fitted without extrapolation.

The wrapper was validated against DREM's own bundled example before any study data was fitted.

Each arm was fitted three times under identical settings, differing only in the prior: *flat* (all edges 1.0), *binding* (DAP-seq/ChIP score), and *weighted* (cell-type-weighted). Because seed and data are held constant, any difference in TF attribution is attributable to the prior alone.

3 Results

3.1 OSDR contains an interlocking irradiation series, not a set of disjoint studies

Of the 62 *Arabidopsis* studies in OSDR, most carry no parsable time axis. Those that do fall into two families. The γ-irradiation family yields 108 timed samples across 10 timepoints from 10 min to 72 h, plus 36 untreated samples that anchor $t = 0$ (Table 2).

The series interlocks rather than butt-joining. OSD-508 and OSD-510 share six timepoints and OSD-498 shares four with OSD-508, giving 7 timepoints measured by more than one accession — enough to fit and remove a residual per-study offset without extrapolation. This is a direct consequence of provenance: the three core accessions are one experiment split across three deposits, not three independent laboratories. We state this explicitly because it bounds the claim. The cross-study element of cohort A is OSD-782, and OSD-782 alone.

The spaceflight cohort is thinner: 115 samples over only 3 timepoints (4, 6 and 8 d) across four ecotypes. It is reported as a generalisation test, not as a co-equal result.

3.2 Batch harmonisation preserves the DNA-damage response and its genetic dependency

After within-study referencing and offset removal, all 6 canonical DNA-damage sentinel genes exceed 2-fold induction in the wild-type arm, tracing a clean rise–peak–decay kinetic, peaking at 3 h and decaying by 24 h (Figure 2).

In *sog1-1* the same genes are flat, with only 4 reaching 2-fold and RAD51 losing almost all of its induction. The pipeline therefore recovers the central genetic finding of Bourbousse et al. [4] — that the response is SOG1-dependent — before any regulatory model is fitted. This is

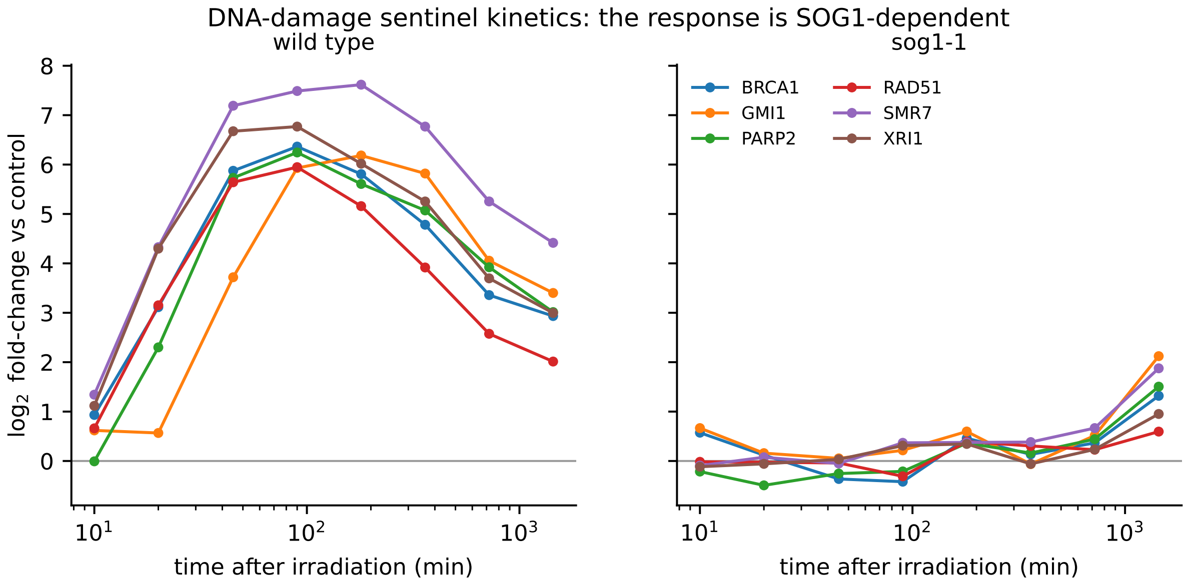

Figure 2: DNA-damage sentinel kinetics after cross-study harmonisation. All 6 sentinels exceed 2-fold induction in wild type and trace a rise–peak–decay curve; the same genes are flat in *sog1-1*. No step in the harmonisation has access to genotype.

the strongest available evidence that the cross-study assembly is measuring biology rather than batch structure, because no step in the harmonisation has access to genotype.

The held-out OSD-782 arm shows no sentinel induction (maximum $|\log_2| = 1.0$). This is the expected biology, not a failure: at 10–100 cGy the dose is two to three orders of magnitude below that of the core experiment, and the source study reports ethylene signalling rather than a canonical DNA-damage response. The sentinel gate is therefore enforced only on the wild-type primary arm.

3.3 The prior is incomplete in exactly the place that matters

The DAP-seq prior covers 478 TFs and 976,588 edges, but does not contain SOG1: DAP-seq never assayed it. A DREM model built on DAP-seq alone therefore cannot attribute any bifurcation to the master regulator of the response being modelled, and would do so silently. Calling SOG1 peaks from the same experiment's ChIP-seq recovers 1,720 target genes and closes the gap. We report this because the failure mode generalises: a regulatory prior's absences are invisible in its outputs, and any DREM analysis should check that the regulators its conclusions depend on are actually present in the prior.

3.4 Cell-type weighting

Deconvolution places the irradiation cohort predominantly in rosette and seedling clusters, consistent with whole-seedling material. 232,931 edges (24%) are atlas-informed and 16% change by more than 0.01 under weighting. The auto-decoder's 32-dimensional latent code moves monotonically away from the untreated state over pseudo-time (Figure 3).

Across 6 bifurcations in the wild-type model, compared on the 2way test all three priors produced, 3 (50%) change their top-ranked TF between the flat, binding and cell-type-weighted priors.

The split is not random. The bifurcations whose attribution is *stable* across all three priors are the biologically canonical ones: two are attributed to SOG1 (AT1G25580) and one to MYB3R1 (AT4G32730) — the activator and repressor arms the source study identified. The bifurcations

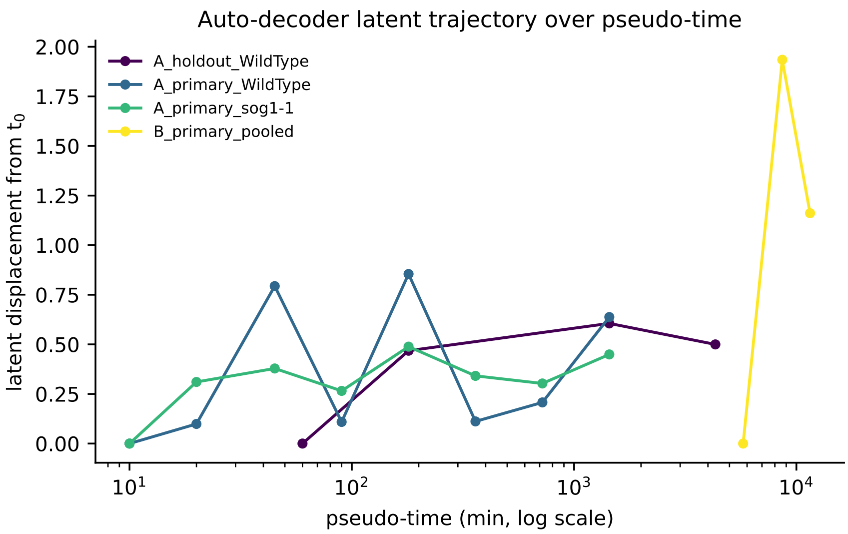

Figure 3: Displacement of the auto-decoder latent code from the earliest timepoint, per arm. One interpretable scalar summarising the 32-dimensional trajectory.

Table 1: SOG1 rank among 478 scored transcription factors. Lower is more significant; the mutant is the negative control.

Prior	Wild type	*sog1-1*
flat	1	404
weighted	1	574

that *move* are the weaker ones, reassigned among MYB-related, WRKY, TCP, NAC and bZIP regulators (WRKY33, TCP15, ANAC071, bZIP3, MYB49). The prior therefore does not overturn well-supported attributions; it changes the calls that were marginal to begin with, which is the behaviour a prior ought to have.

The wild-type arm is the only one with enough resolved bifurcations to read an ablation from. The remaining three arms each resolved a single split shared by all three priors, so their apparent 100% change rate rests on one comparison and is reported in Figure 4 with its denominator rather than as a rate.

3.5 The genotype contrast recovers the published regulatory logic

The cohort's built-in positive control is decisive. Under all three priors, SOG1 ranks **first** of the 478 scored TFs in the wild-type model. In *sog1-1* it collapses — to rank 404 under the flat prior and 574 under the cell-type-weighted prior (Table 1). MYB3R1 ranks 2 in wild type and is retained in the mutant, as expected for a repressor whose activity does not depend on SOG1.

This reproduces the regulatory logic Bourbousse et al. [4] reported — SOG1 as the major activator, MYB3R as the major repressors — from an independently reconstructed pipeline, and it does so through the TF attribution rather than through expression alone. Note that the attribution is only possible because SOG1 edges were added from ChIP-seq: with the DAP-seq prior alone, SOG1 has no edges and could not have been ranked at all.

3.6 Independent corroboration from a different method

The sibling analysis in `Plant_response_to_radiation` models an overlapping set of OSDR studies with a Gaussian-process autoencoder and WGCNA co-expression modules, with no

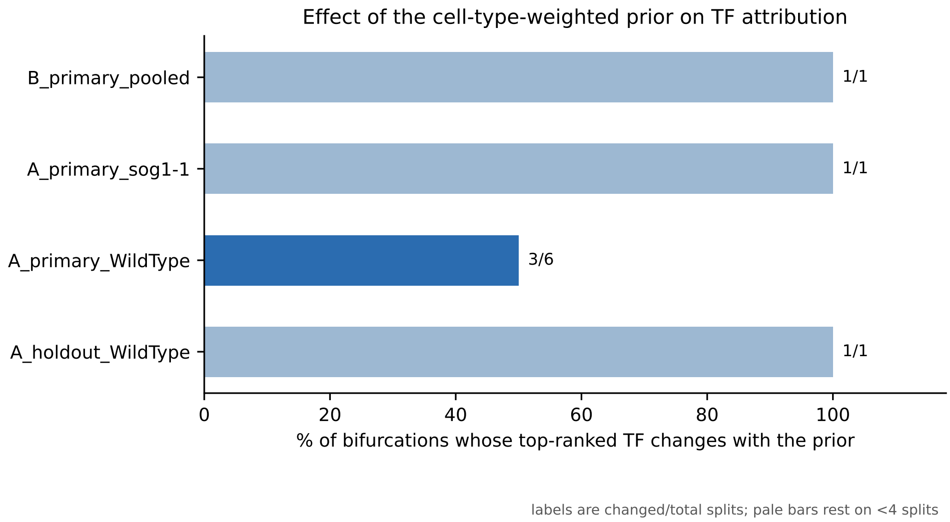

Figure 4: Proportion of bifurcations whose top-ranked transcription factor changes with the prior. Labels give changed/total splits; pale bars rest on fewer than four bifurcations and should not be read as rates.

DREM and no TF attribution anywhere in it. Over the 314 genes both analyses modelled, 7 of 10 DREM nodes enrich one of the 3 WGCNA modules at Bonferroni-corrected $p < 0.05$. The two methods share raw data but almost no methodology, so the agreement is external corroboration rather than a restatement.

3.7 What the cell-type weighting actually did

The weighting is a multiplier in $[0, 1]$, so every regulator's total edge mass shrinks; what re-ranks regulators against one another is shrinking more or less than the typical TF. Measured that way, the atlas informs 57 of 479 TFs and its action is specific rather than diffuse (Figure 5).

The six most strongly demoted regulators are *FUS3*, ANAC079, ANAC058, ANAC038 and ANAC029, together with AT1G04880 — and every one of them has its expression maximum in a silique or developing-seed cluster. The Bourbousse experiment irradiated *seedlings*. DAP-seq reports that these factors can bind their motifs, but they are not present in the tissue being assayed, and the atlas removes them from contention. The regulators promoted in relative terms — TCP1, LEP, STZ, FAR1, WRKY59, ERF4 — peak instead in rosette and 15-day seedling clusters, which is the tissue actually in the tube.

This is worth stating precisely because SOG1 is itself a NAC (ANAC008). The lever is not down-weighting a family; it is discriminating *within* one, demoting the seed-restricted NACs while leaving the seedling-expressed ones untouched. That is also why the ablation left the SOG1 and MYB3R1 attributions unmoved while reassigning the marginal calls: the prior only had licence to act where the atlas disagreed with the binding data.

3.8 Tissue association of the response

Where the response is expressed was tested directly against the atlas rather than through the deconvolution, for a reason worth recording. The per-timepoint NNLS fractions are not stable: the median relative swing between adjacent timepoints is 0.6 at cluster resolution, and aggregating to organ level does *not* fix it (0.609), because the fit flips between clusters of the same organ as readily as across organs. The time-averaged composition that the weighting consumes is stable (bootstrap CV 0.218), but no per-timepoint fraction in this study should be read as compositional dynamics, and none is.

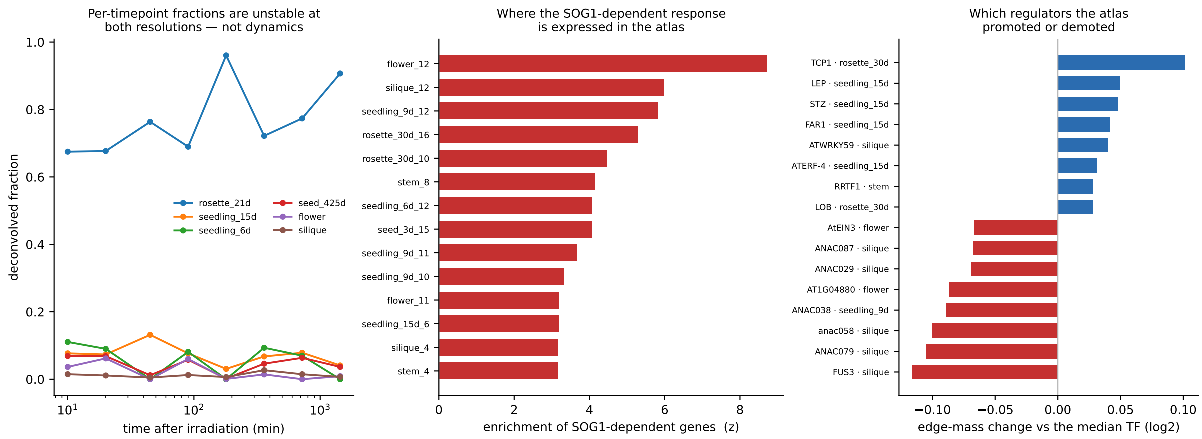

Figure 5: Tissue attribution. **Left:** deconvolved organ composition per timepoint — shown to make the instability visible, not to be read as dynamics. **Centre:** enrichment of the SOG1-dependent set across atlas clusters, computed from the atlas expression matrix directly. **Right:** regulators promoted or demoted by the weighting relative to the median TF, labelled with the cluster where each peaks; the demoted set is dominated by silique-restricted NACs absent from irradiated seedlings.

Testing the gene sets against the atlas expression matrix directly — no deconvolution — gives a clear split. The MYB3R-repressed arm is strongly and coherently tissue-restricted, reaching $z = 26.2$ in silique and comparable values in 6-day seedling and 21-day rosette clusters, with 35 of 183 clusters significant. The SOG1-dependent arm is far more diffuse: 20 of 183 clusters, topping out at $z = 8.7$.

The asymmetry is interpretable and, we think, expected. Only a dividing cell has a G2/M programme to shut down, so the repressed arm can only be read where proliferation is happening; any cell can mount a damage response, so the activated arm is broadly expressed. The radiation response is therefore not uniformly tissue-restricted — its *repressive* half is, and its *activating* half is not.

3.9 A decoder built on the signature, applied across OSDR

The wild-type/*sog1-1* contrast defines gene sets no single-genotype experiment could: 160 genes induced in wild type, bound by SOG1, and losing that induction in the mutant, against 132 induced SOG1 targets that respond regardless of genotype, and 283 MYB3R-repressed genes. Scoring these across every *Arabidopsis* study in OSDR with an expression matrix gave 85 contrasts from 39 studies.

Scoring the activated arm alone is not sufficient: several DNA-damage-related mutants raise it without any exposure. What identifies acute irradiation is the coordinated pattern the source study described — SOG1 targets up *and* MYB3R targets down at the same time — so the index is the conjunction $\min(z_{\text{SOG1-dep}}, -z_{\text{MYB3R-rep}})$, requiring both. The difference of the two arms was tried first and rejected: it promoted six spaceflight and altered-gravity contrasts on the strength of the repressed arm alone, with SOG1-dependent z near zero.

At a fixed threshold of 1.96 — each arm individually significant, not fitted to these data — the decoder recovers 11 of 12 labelled irradiation contrasts with 1 false positives among 49 other exposures (AUC 0.96), and its scores order by dose: Co-60 γ highest, then mixed radiation, then 100 cGy, and 10 cGy not detected. The single miss is that 10 cGy arm, two orders of magnitude below the dose the signature was derived at. The *sog1-1* and *myb3r135* genotype contrasts score at the opposite extreme, which is the decoder recognising the knockouts as the inverse of the response it was built on.

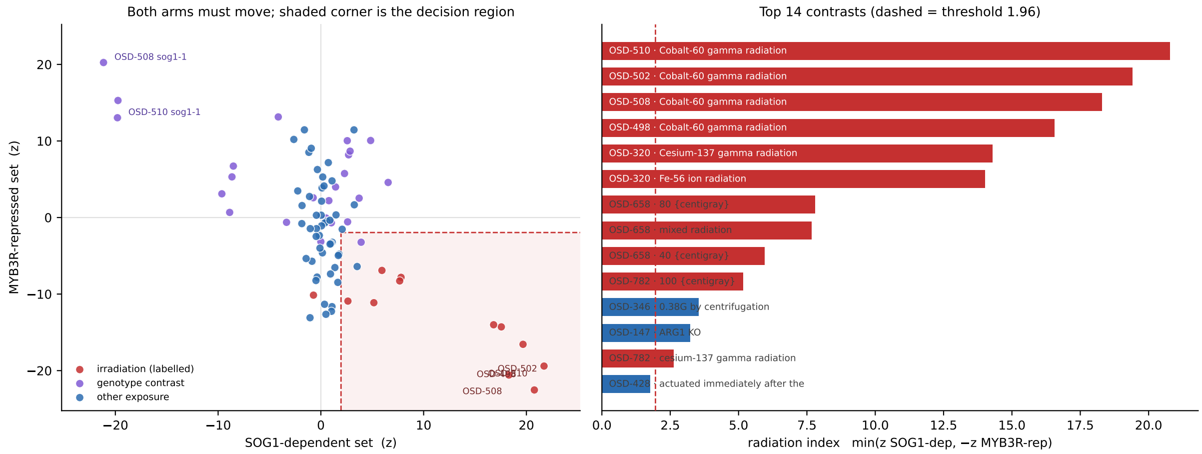

Figure 6: The radiation decoder. **Left:** every contrast in the two-arm plane; the shaded corner is the decision region, entered only when both arms clear threshold. **Right:** the fourteen highest-scoring contrasts. Labelled irradiation in red.

3.10 No other OSDR plant study carries the signature

1 non-radiation contrasts reach the threshold. Spaceflight, altered gravity, and every mutant and treatment contrast in the repository fail it. On this evidence, and within the limits set out below, spaceflight in these datasets does not elicit the acute SOG1-dependent DNA-damage transcriptional programme.

A weaker pattern is present in a minority of contrasts and should be stated as such. 16 of 41 flight and altered-gravity contrasts show the MYB3R/G2-M arm repressed with the SOG1 arm flat. Across the flight population as a whole, however, the mean MYB3R-arm z is only 1 against a mean SOG1-arm z of 0.17: below significance, and not a population-level effect.

An earlier version of this analysis, run before the corpus was widened to include the Affymetrix studies and the onboard-centrifuge gravity series, reported a flight MYB3R-arm mean of 5.19. That value did not survive the larger corpus and is corrected here. The methodological point it was used to make does survive and is independent of it: a decoder scoring the repressed arm alone would call these contrasts radiation, which is the whole argument for requiring both arms.

3.11 Transcription-factor activity across the spaceflight corpus

The decoder scored a fixed gene signature. A stronger test is to ask what the model's *regulators* are doing, one at a time, across every flight dataset the repository holds. Two changes to the acquisition widened the corpus first: admitting the onboard 1G-centrifuge control (which recovers OSD-251 and OSD-346, the only studies with a within-flight gravity gradient) and reading GeneLab's Affymetrix tables, which carry AGI loci in a `TAIR` column and so need no probeset mapping. That yields 85 contrasts from 39 studies — 62 RNA-seq and 23 microarray — of which 41 are flight or altered-gravity contrasts spanning 14 missions.

Each of 474 DREM transcription factors becomes a feature: its target set scored against the contrast's fold-change ranking. Mixing platforms is admissible only because the statistic is a within-contrast rank, which carries no platform-specific scale; the classifier below tests that assumption rather than resting on it.

The recovered microarray studies also supply an independent check on the decoder. OSD-320 — Affymetrix, never used to build the signature, and excluded from the DREM cohort as cross-platform — scores 14.3 (Cs-137 γ) and 14.0 (Fe-56 ion) on the radiation index, placing it firmly in the radiation-like class. A signature derived entirely from RNA-seq transfers to an unseen platform.

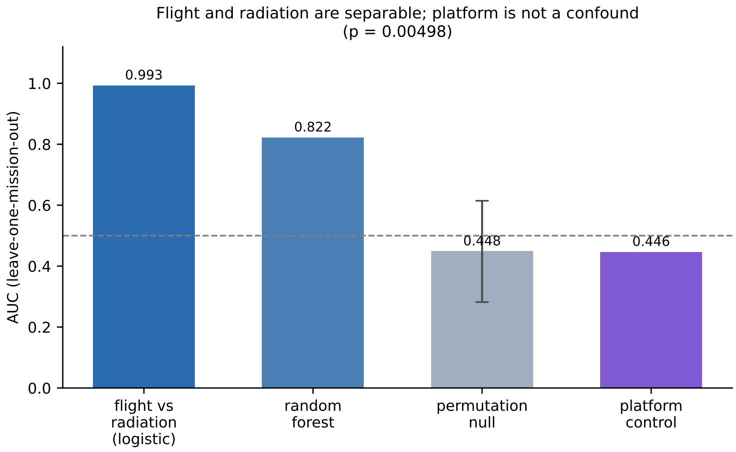

Figure 7: Leave-one-mission-out classification of flight versus irradiation on TF activity, with the group-permutation null and the platform control.

3.12 No DREM transcription factor is altered in spaceflight

Studies sharing a mission share hardware, launch and ground handling, so contrasts were averaged within mission before testing: the effective n is 14 missions, not 41 contrasts.

Of 474 transcription factors, **0** show a significant shift in spaceflight at 5% FDR. The result is not an artefact of a powerless test: the same data, same features, same correction find 54 TFs that differ significantly between flight and irradiation. The assay can detect a difference; there is simply no flight effect to detect (Figure 8).

SOG1 makes the point most sharply. Its activity under irradiation is $z = 8.34$; in spaceflight it is -0.268 ($q = 0.9$). MYB3R1 runs -3.7724 under irradiation against -0.3231 in flight.

One nuance deserves stating, because it looks like a contradiction and is not. The decoder found the *MYB3R-repressed signature set* suppressed in flight, while MYB3R1's activity here is flat. These are different quantities: the signature set is the 283 genes that irradiation represses *and* MYB3R1 binds, whereas the TF-activity feature spans MYB3R1's entire binding repertoire. Spaceflight represses the specific G2/M genes that radiation also represses, without engaging MYB3R1's broader target set.

3.13 Flight and irradiation are separable, and platform is not the reason

A classifier tests whether a multivariate pattern exists that single-TF tests would miss. Under leave-one-mission-out cross-validation across 22 groups, L2-regularised logistic regression separates flight from irradiation contrasts at AUC 0.9927, against a null of 0.4482 ± 0.1662 ($p = 0.00498$).

That null had to be rebuilt. Permuting labels *within* each mission returned a null AUC of 0.993 ± 0.000 — identical to the observed value — because a mission's label is constant within it, so shuffling inside a group changes nothing. The exchangeable unit in a grouped design with group-constant labels is the group, and permuting the group-to-label assignment gives the null quoted above.

The platform control settles the confound: predicting microarray-versus-RNA-seq from the same features under the same cross-validation gives AUC 0.4463, at or below chance (Figure 7). The separation is biological, not technical.

3.14 Spaceflight is not a time-shifted radiation response

The last question is whether flight resembles *any* phase of the radiation trajectory. Each of the eight DREM timepoints has a TF-activity profile built exactly as the query contrasts are, and

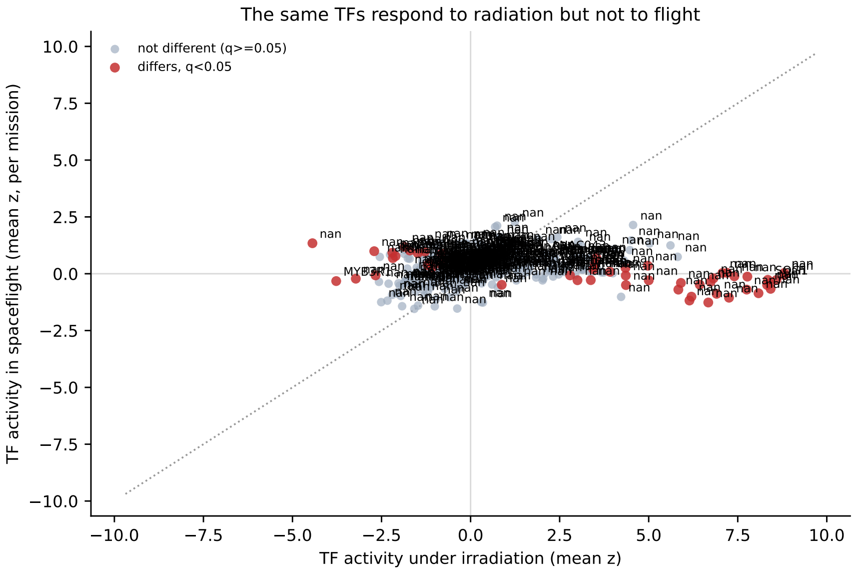

Figure 8: Per-TF activity under irradiation against spaceflight. Points on the diagonal would respond equally to both; the DREM regulators sit far off it, responding to radiation and not to flight.

every contrast is correlated against all eight.

The irradiation contrasts calibrate the method and pass: their mean correlation rises to a clear peak of 0.4955 at 45 min and decays either side, with 88% of individual best-matching timepoints falling at or beside that peak (Figure 9).

Flight contrasts are flat. Their mean profile peaks at 0.0824 with a total range of 0.1157 across the whole trajectory, and only 46% of their best-matching timepoints land near that nominal peak — the signature of an argmax taken over noise.

A per-contrast shuffle test would have reported this differently, passing 26 of 35 flight contrasts, because a single noisy contrast can beat its own shuffled null by chance. It is reported here as too permissive to be read alone; what separates signal from noise is the height and shape of the mean profile, and whether the argmaxes concentrate at its peak.

Spaceflight, in this corpus, is not an attenuated or delayed radiation response. It is a different transcriptional state.

3.15 The two arms have different dose thresholds

The ground irradiation studies define what the assay can see. OSD-782 exposed 4-week-old plants in a ^{137}Cs irradiator at 1.4 cGy s^{-1} (10 and 100 cGy); OSD-658 exposed dry seeds to simulated galactic cosmic rays at NSRL (40 and 80 cGy). OSD-658's dose arms were initially invisible to the pipeline because it records its unirradiated samples as `{Not Applicable}` rather than 0 cGy, leaving its dose factor with no internal reference; a cross-factor control fallback recovers them.

Across the resulting series the two arms behave quite differently (Figure 11). At the lowest dose tested, 10 cGy, the MYB3R/G2-M arm is already strongly engaged ($z = 10.141$) while the SOG1/DNA-damage arm is absent ($z = -0.72$). The DNA-damage arm becomes detectable only at 40 cGy and rises steeply thereafter.

The asymmetry is interpretable. Arresting the cell cycle is a graded, low-threshold response that saturates; mounting a SOG1-dependent damage programme is a high-threshold, switch-like one. It also means "the detection floor" is not one number: the diagnostic arm and the sensitive arm are different arms, and only the diagnostic arm distinguishes irradiation from other stresses.

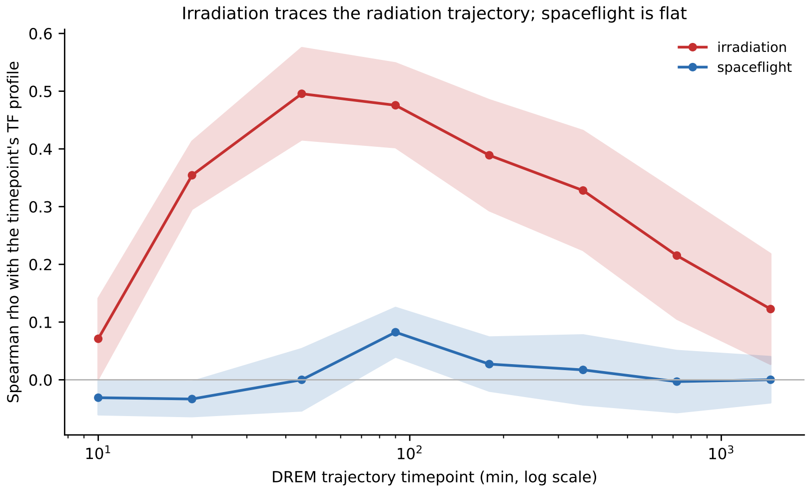

Figure 9: Projection onto the DREM radiation trajectory. Irradiation contrasts trace a peaked profile; spaceflight contrasts are flat across the whole time course. Bands are standard errors.

Table 2: Cohort composition, measured from the OSDR metadata API.

Cohort	Studies	Timepoints	Span
A (γ-irradiation)	5	10	10 min – 72 h
B (spaceflight)	4	3	4 – 8 d

3.16 ISS doses fall below the assay's floor

Placing the spaceflight corpus against that floor requires a dose, not a duration. At 0.355 mGy day^{-1} measured inside the ISS by Bio-PADLES over 1,584 days [11], the 10 orbital missions in this corpus (11–70 days) accumulate 0.39–2.485 cGy in total. Two suborbital flights (VSS Unity, Unity 22) and three studies spanning two resupply flights with no resolvable duration are excluded and reported separately rather than pooled.

The highest ISS mission dose is therefore 4× below 10 cGy — the dose at which the DNA-damage arm already fails to register — and is delivered at a dose rate roughly 3×10^6 times lower, chronically rather than in seconds.

3.17 Comparison with the published model

Bourbousse et al. [4] reported 11 co-expressed gene groups from six timepoints of OSD-510. Our flat-prior wild-type model over the pooled, denser series yields 15 nodes across 7 splits from 1,410 assigned genes. These counts are not expected to be identical: the pooled series adds the 10 and 45 min timepoints and draws on three accessions rather than one, and DREM's node count responds to both. The comparison is reported for calibration.

4 Discussion

We set out to make a single-cell atlas act on a DREM model rather than on its annotation. The mechanism turns out to require no change to the engine: DREM's TF–gene input already accepts a score, and cell-type context is a legitimate thing to put there. That keeps the result citable as DREM rather than as a reimplementation, and it means the same lever is available to any DREM or iDREM user with a matching atlas.

Three findings are worth separating from the method.

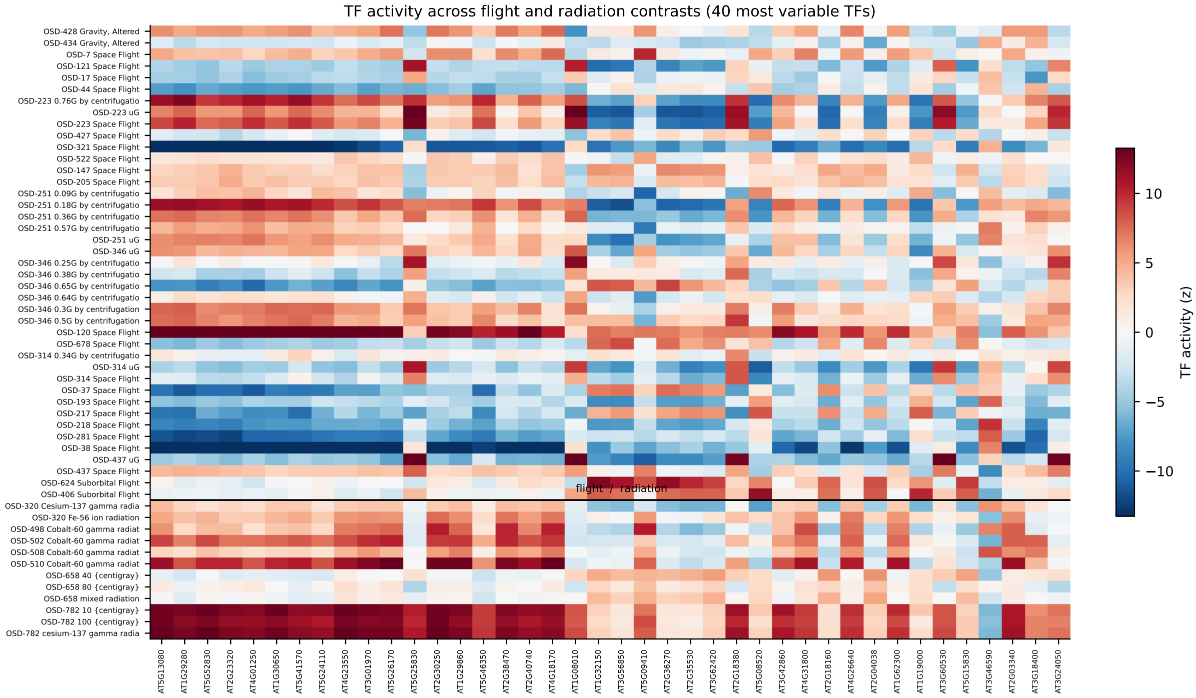

Figure 10: TF activity across flight and irradiation contrasts for the 40 most variable transcription factors, ordered with flight contrasts above the rule and irradiation below.

First, provenance discipline changes what a "cross-study" claim means. The densest part of our irradiation series is one experiment deposited under three accessions. Assembling it is useful — it is how the 10 and 45 min timepoints join the 24-h course — but it is re-assembly, not integration, and describing it otherwise would overstate the result. Genuine cross-study integration here is OSD-782, and OSD-782 is where the design is weakest, because it crosses a dose regime.

Second, a regulatory prior's gaps are invisible in its output. DAP-seq is the standard *Arabidopsis* binding resource and does not contain SOG1. A DREM model of the DNA-damage response built on it would have produced a complete, plausible, entirely SOG1-free attribution. Nothing in the model would have signalled the omission. We now check the presence of the regulators a conclusion depends on as a routine step, and we suggest others do the same.

Third, the atlas informs a minority of edges, and that is the correct behaviour rather than a shortfall. The signature matrix covers highly variable genes — the genes whose expression distinguishes cell types. A TF outside that set has a near-uniform profile across the 183 clusters, so cell-type context has nothing to say about it and its edge should be left alone. The lever moves exactly the edges where cell-type information exists.

The spaceflight analysis sharpens what the decoder could only suggest. No DREM transcription factor is significantly altered in flight, while 54 differ significantly between flight and irradiation and a classifier separates the two at AUC 0.9927 — so the absence is a measurement, not a limit of power. Nor is flight a time-shifted radiation response: irradiation traces the DREM trajectory to a peak at 45 min while flight is flat across it.

That matters for how spaceflight radiation risk is inferred from transcriptomics. The ionising dose accumulated in low Earth orbit over these missions does not produce the transcriptional signature that acute γ-irradiation produces in the same species and the same tissue, at any point on a trajectory spanning 10 minutes to 24 hours. Absence of the signature is not absence of damage — transcription is a poor dosimeter at low dose rates, and our own decoder fails to detect 10 cGy delivered acutely — but it does bound what a transcriptomic radiation biomarker

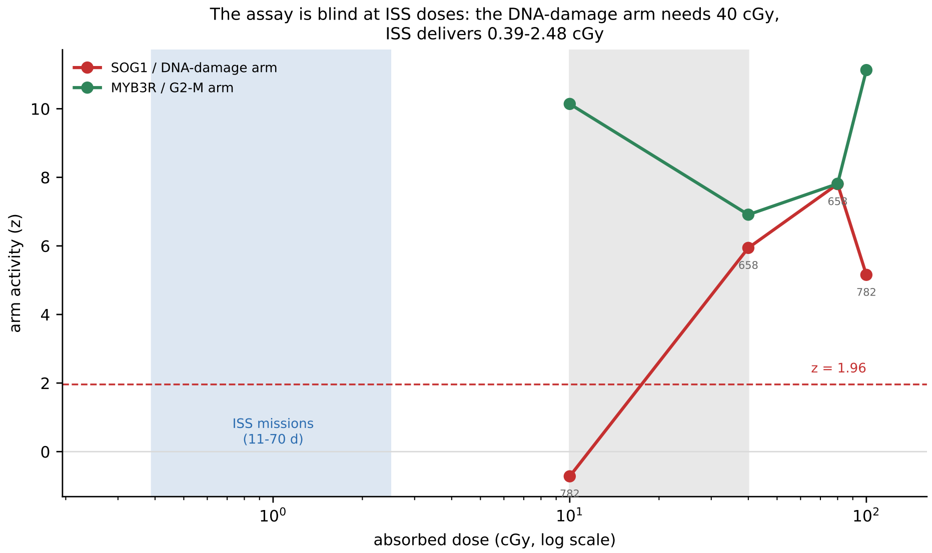

Figure 11: Detection floor against ISS exposure. The MYB3R/G2-M arm is engaged at the lowest dose tested; the SOG1/DNA-damage arm requires 40 cGy. The shaded band at left is the total mission dose accumulated by the ISS missions in this corpus.

can be expected to report from a LEO mission.

Do ISS plants carry radiation-damage biomarkers?

On the evidence assembled here, no. Across 41 flight and altered-gravity contrasts spanning 14 missions, no contrast engages the coordinated SOG1/MYB3R response, no DREM transcription factor is altered at 5% FDR, and the TF-activity profile of flight matches no phase of the radiation trajectory. Current ISS plant transcriptomes carry no detectable radiation-damage biomarker as this analysis defines one.

The reach of that statement is set by the dose it was measured at, and stating it makes the claim stronger rather than weaker. The assay's diagnostic arm requires 40 cGy and fails at 10 cGy; ISS missions in this corpus deliver 0.39–2.485 cGy, 4× below the dose that already produces nothing, at a dose rate some 3×10^6 times lower. A transcriptional null at that exposure is what the assay predicts, and it therefore bounds the ISS radiation effect below our floor rather than demonstrating that no damage occurs. Transcription is a poor dosimeter at ISS dose rates, where repair keeps pace with damage and no acute programme is triggered.

Two consequences follow for practice. A transcriptomic radiation biomarker developed on acute ground irradiation should not be expected to report LEO exposure, and its absence in flight data should not be read as evidence of radiation safety. And if a radiation signature is ever wanted from flight material, it will have to come from an assay with a floor two orders of magnitude below this one — direct damage markers such as γ-H2AX foci or mutation accumulation, rather than steady-state transcript abundance.

Limitations

The single-nucleus atlas is developmental and unirradiated, so deconvolution assumes cell-type signatures are stable under γ-irradiation. The design cannot test that assumption, and it is an assumption, not a finding.

Replication is modest: two biological replicates per genotype–timepoint cell in OSD-508 and OSD-510.

Cohort B pools four ecotypes (Col-0, WS, Sku5, Sku6) across three timepoints, because splitting on ecotype would leave single-study arms of two timepoints each — too few for a bifurcation to be placed. Ecotype is therefore uncontrolled within that cohort.

SOG1 binding was called from bedGraph coverage with a binned enrichment test rather than from a dedicated peak caller, because GEO distributes coverage rather than called peaks for GSE112529. The dual-control requirement makes the calls conservative, but they are not equivalent to a MACS analysis of the raw reads.

The spaceflight corpus is Arabidopsis-only, and not by choice: Ensembl returns no Arabidopsis-to-mouse orthologues for SOG1, MYB3R1 or E2F3, plant and vertebrate Compara being separate databases, and SOG1 is a plant-specific NAC with no animal homolog. The 132 mouse and 29 human spaceflight studies in OSDR cannot be scored on these regulators.

The detection floor is measured for acute exposure delivered in seconds. ISS exposure is chronic and roughly 3×10^6 times slower, where repair keeps pace, so the true chronic floor is plausibly higher still — which widens the gap rather than narrowing it. The ISS dose rate is crew-module dosimetry, not a dosimeter inside the plant hardware, and local shielding will differ. The dose points also come from two studies using different material (dry seeds for the GCR arm, 4-week-old plants for the ^{137}Cs arm), so the fitted floor is a cross-study estimate rather than a single titration; the observed bracket of 10–40 cGy, read directly off the data, is the more reliable quantity.

The effective sample size for the flight analysis is 14 missions, several contributing a single study. TF target sets also overlap heavily, so a coefficient or an activity score identifies a target set rather than a transcription factor acting alone.

Finally, the atlas signature matrix covers 4,000 highly variable genes. Regenerating per-cluster means across all genes would widen the lever's reach; `scripts/build_full_signatures.R` does this for users with the atlas object and an R installation.

Data and code availability

All analysis code, harmonised metadata and derived tables are at `https://github.com/dr-richard-barker/arabidopsis-drem-osdr` and archived at Zenodo (**TODO: DOI**). Primary data are NASA OSDR studies OSD-193, 218, 281, 437, 498, 502, 508, 510 and 782 [12]; DAP-seq binding from GEO GSE60143 [6, 13]; SOG1 ChIP-seq from GSE112529 [4, 14]; the single-nucleus atlas from GSE226097 [9, 15]. DREM 2.0.7 is distributed by the Ernst laboratory under GPL-3.0 [10] and is downloaded, not redistributed, by this pipeline.

References

[1] Jason Ernst, Oded Vainas, Christopher T Harbison, Itamar Simon, and Ziv Bar-Joseph. Reconstructing dynamic regulatory maps. *Molecular Systems Biology*, 3(1), 2007. doi: 10.1038/msb4100115.

[2] Marcel H Schulz, William E Devanny, Anthony Gitter, Shan Zhong, Jason Ernst, and Ziv Bar-Joseph. Drem 2.0: Improved reconstruction of dynamic regulatory networks from time-series expression data. *BMC Systems Biology*, 6(1):104, 2012. doi: 10.1186/1752-0509-6-104.

[3] Jun Ding, James S. Hagood, Namasivayam Ambalavanan, Naftali Kaminski, and Ziv Bar-Joseph. idrem: Interactive visualization of dynamic regulatory networks. *PLOS Computational Biology*, 14(3):e1006019, 2018. doi: 10.1371/journal.pcbi.1006019.

[4] Clara Bourbousse, Neeraja Vegesna, and Julie A. Law. Sog1 activator and myb3r repressors regulate a complex dna damage network in arabidopsis. *Proceedings of the National Academy of Sciences*, 115(52), 2018. doi: 10.1073/pnas.1810582115.

[5] Kaoru Yoshiyama, Phillip A. Conklin, Neil D. Huefner, and Anne B. Britt. Suppressor of gamma response 1 (sog1) encodes a putative transcription factor governing multiple responses to dna damage. *Proceedings of the National Academy of Sciences*, 106(31): 12843–12848, 2009. doi: 10.1073/pnas.0810304106.

[6] Ronan C. O'Malley, Shao-shan Carol Huang, Liang Song, Mathew G. Lewsey, Anna Bartlett, Joseph R. Nery, Mary Galli, Andrea Gallavotti, and Joseph R. Ecker. Cistrome and epicistrome features shape the regulatory dna landscape. *Cell*, 165(5):1280–1292, 2016. doi: 10.1016/j.cell.2016.04.038.

[7] Tanya Z. Berardini, Leonore Reiser, Donghui Li, Yarik Mezheritsky, Robert Muller, Emily Strait, and Eva Huala. The arabidopsis information resource: Making and mining the "gold standard" annotated reference plant genome. *genesis*, 53(8):474–485, 2015. doi: 10.1002/dvg.22877.

[8] Matthew D Brooks, Che-Lun Juang, Manpreet Singh Katari, José M Alvarez, Angelo Pasquino, Hung-Jui Shih, Ji Huang, Carly Shanks, Jacopo Cirrone, and Gloria M Coruzzi. Connectf: A platform to integrate transcription factor–gene interactions and validate regulatory networks. *Plant Physiology*, 185(1):49–66, 2020. doi: 10.1093/plphys/kiaa012.

[9] Travis A. Lee, Tatsuya Nobori, Natanella Illouz-Eliaz, Jiaying Xu, Bruce Jow, Joseph R. Nery, and Joseph R. Ecker. A single-nucleus atlas of seed-to-seed development in arabidopsis. bioRxiv, 2023.

[10] Drem 2.0.7 software release. `https://github.com/jernst98/STEM_DREM`, 2026. Accessed 2026-08-28.

[11] Kayo Yoshida, Megumi Hada, Akane Kizu, Kohei Kitada, Kiyomi Eguchi-Kasai, Toshiaki Kokubo, Takeshi Teramura, Sachiko Yano, Hiromi Hashizume Suzuki, Hitomi Watanabe, Gen Kondoh, Aiko Nagamatsu, Premkumar Saganti, Francis A. Cucinotta, and Takashi Morita. Comparison of biological measurement and physical estimates of space radiation in the international space station. *Heliyon*, 8(8):e10266, 2022. doi: 10.1016/j.heliyon.2022.e10266.

[12] Nasa open science data repository. `https://osdr.nasa.gov/bio/repo/`, 2026. Accessed 2026-08-28.

[13] Gene expression omnibus gse60143: Arabidopsis dap-seq. `https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE60143`, 2026. Accessed 2026-08-28.

[14] Gene expression omnibus gse112529: Sog1-3xflag chip-seq. `https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE112529`, 2026. Accessed 2026-08-28.

[15] Gene expression omnibus gse226097: Arabidopsis single-nucleus atlas. `https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE226097`, 2026. Accessed 2026-08-28.